— PORTFOLIO CASE STUDY

V2.0

AI Part Prioritization Framework

Systemizing sourcing triage under ambiguity

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Stack: Excel + Claude AI

01 — PROBLEM

If everything is important, what is actually urgent?

Sourcing engineers in complex manufacturing are routinely handed portfolios of 50 to 100 parts and asked to drive impact. There is rarely a structured method to determine where to allocate effort, which risks are existential versus noise, or which cost opportunities represent real leverage.

Prioritization defaults to reactive, meeting-driven, and personality-influenced, not system-driven.

> "Why are you working on that part first?" A question that rarely had a data-backed answer.

02 — INSIGHT

Production continuity first. Not spend alone.

A $50K part that stops a line is more urgent than a $2M part with diversified supply and stable delivery. The right system must be explainable, surface operational risk, and produce defensible action categories without relying on escalation pressure or institutional memory.

Supply continuity failures in aerospace are catastrophic and difficult to reverse. Cost opportunity, while meaningful, is a second-order priority.

03 — APPROACH

A weighted scoring engine across four dimensions

PRIORITY SCORE FORMULA

Score = ((Production Impact × 0.35) + (Supply Risk × 0.30) + (Cost Exposure × 0.20) + (Instability × 0.15)) / 5 × 100

Weights are configurable. Production First (Default): 35/30/20/15 | Cost Down: 15/20/40/25 | Supply Resilience: 25/40/20/15 | Custom: user-defined

35%

PRODUCTION IMPACT

— CTB status
— Line-stop risk
— Safety stock (days)

30%

SUPPLY RISK

— Single-source exposure
— Supplier count
— Geographic concentration
— Special process requirements

20%

COST EXPOSURE

— Annual spend
— Should-cost delta (%)

15%

INSTABILITY

— On-time delivery %
— Quality issues
— Chronic open status

Default weights reflect a production-continuity-first philosophy. All weights are configurable in the SETTINGS tab. Switch modes to reprioritize the portfolio based on current business objective without rebuilding the model.

04 — V2 UPGRADES

Four new capabilities built on the V1 foundation

CAPABILITY	DESCRIPTION
Dynamic Weight Configuration	A dedicated SETTINGS tab allows users to switch between three preset scoring modes: Production First, Cost Down, and Supply Resilience, or define custom weights. The entire ranking updates instantly without touching any formulas.
Reason Codes	Each ranked part now displays a plain-language reason code in the OUTPUT tab explaining why it scored where it did. Example: "CTB, Single Source, OTD Critical, High Cost

CAPABILITY	DESCRIPTION
	Delta." Eliminates the need to scan across eight columns in leadership meetings.
Should-Cost Flag	Parts missing should-cost delta data are flagged with a warning in the OUTPUT tab. Surfaces incomplete data before sourcing action is taken rather than silently treating missing values as zero.
Risk vs. Opportunity Matrix	A dedicated MATRIX tab plots all 60 parts across Supply Risk (X) and Cost Exposure (Y). Four quadrants — Act Now, Cost Opportunity, Risk Watch, and Monitor — give an instant visual overview of portfolio composition and sourcing priorities.

05 — FRAMEWORK CONTROLS

Three features that make the model ops-ready for day-to-day sourcing workflows

FEATURE	DESCRIPTION
Data Confidence Flag	Each part is rated H/M/L for input reliability. Low-confidence parts surface a warning before action is taken. The tool informs, the sourcing manager decides.
Escalation Override	Leadership-escalated parts receive a minimum score of 70 and forced action category without corrupting the underlying calculated score. Political context stays visible without contaminating model integrity.
AI Rationale Layer	Top 10 ranked parts are passed through a structured Claude prompt generating a 3-sentence business rationale and action recommendation in plain language, ready for leadership reviews. AI does not create scores or facts — it only rewrites computed drivers into a narrative. Scoring and action categories are fully deterministic.

From qualitative noise to structured tradeoffs

Instead of "Part A feels urgent," the system produces a scored, ranked, and explainable output with reason codes automatically generated from the underlying data:

P-1040

High-Temp Injector Plate (synthetic)

90
/ 100

DIMENSION SCORES

PRODUCTION IMPACT	SUPPLY RISK	COST EXPOSURE	INSTABILITY
5.0	**4.0**	**4.0**	**5.0**

ACTION CATEGORY

CATEGORY **ESCALATE — Supplier Crisis + Production Risk**

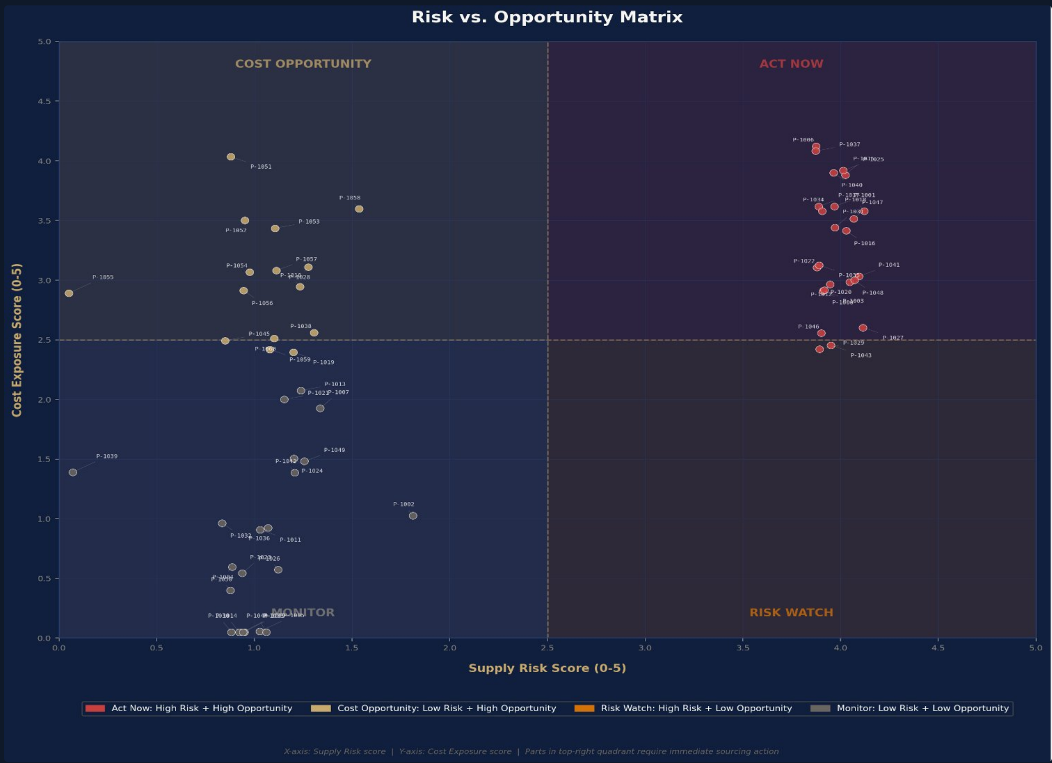
CTB · Single Source · OTD Critical · High Cost Delta · $1M+ Spend · Low Safety Stock

Immediate action: dual-source qualification and cost recovery negotiation.

All parts, part numbers, and values shown above are synthetic and illustrative only.

OUTPUT Tab — 60-Part Ranked Priority List with Reason Codes

MATRIX TAB — RISK VS. OPPORTUNITY MATRIX



X-axis: Supply Risk score | Y-axis: Cost Exposure score | Parts in top-right quadrant require immediate sourcing action

Score profile drives action, not total score alone

The same overall score can result from very different underlying conditions. Action categories are assigned by profile. Supplier Crisis takes precedence over Dual Source escalation when both conditions are met.

CONDITION	ACTION CATEGORY
Leadership Escalation = Y	**ESCALATE — Active Leadership**
Production Impact 4+ and Instability 4+	**ESCALATE — Supplier Crisis + Production Risk**
Production Impact 4+ and Supply Risk 3.5+	**ESCALATE — Dual Source + Eng Review**
Production Impact 4+	**Dual Source Priority**
Supply Risk 3+ and Cost Exposure 3+	**Strategic RFQ — Cost Down + Risk**
Supply Risk 4+	**Supply Risk Mitigation**
Cost Exposure 4+	**Cost Down / RFQ**
Instability 3+	**Supplier Performance Review**
Priority Score 40 or below	**Monitor**

08 — BROADER CONTEXT

One tool in a sourcing operations suite

This framework is the first in a series of structured sourcing tools being built to address the most common pain points in manufacturing procurement. Each tool is designed to stand alone, solve a specific operational problem, and produce outputs that are defensible in a leadership setting.

SUITE PROGRESS

- → Part Prioritization Framework — Complete
- → RFQ Lifecycle Tracker — Complete
- → Parametric Should-Cost Model V2 — Complete
- → Supplier Performance & Risk Scorecard — In Progress
- → Make vs. Buy Decision Framework — Planned

TOOL DESIGN PHILOSOPHY

Each tool is designed to stand alone, solve a specific operational problem, and produce outputs that are defensible in a leadership setting. Structured. Explainable. Action-oriented.

09 — CLOSING NOTE

Built to structure sourcing decisions under pressure. Intentionally simple in structure, rigorous in logic. The goal is clarity, repeatability, and leverage — not complexity. **The model surfaces the data. The sourcing manager makes the call.**

Excel-native decision support tool, built iteratively across two versions. | 60-part synthetic dataset, structure validated against real workflow constraints. | 7-tab workbook. | Production-continuity-first philosophy.